Multi-Agent Scientific Paper Review

Powered by Phi-3 Mini · May 03, 2026 20:04

File	1-s2.0-S092523122300440X-main (2).pdf
Words	13,647
Sections Detected	preamble, methodology, literature, keywords, introduction, results, conclusions, experiments, discussion, r
Review Mode	FAST
Model	qwen3:8b

■ *AI-ASSISTED REVIEW NOTICE: This report was generated by an artificial intelligence system using Phi-3 Mini as the language model. It is intended as a supplementary tool to assist human peer reviewers. All findings must be validated by a qualified domain expert before editorial decisions are made. The AI may miss context-specific nuances and should not be used as the sole basis for acceptance or rejection.*

Parser Warnings

- Abstract not detected as separate section — content included in full text.

Editorial Decision

REJECT

Weighted Score: 2.3 / 5.0
Confidence: 74%

Evaluation Score Table

Scale: 0 = Not evaluable | 1 = Very weak | 2 = Weak | 3 = Acceptable | 4 = Good | 5 = Excellent

Criterion	Score	Rating	Visual
Originality & Novelty	**2.5**	Weak	■■■■■
Manuscript Structure	**2.0**	Weak	■■■■■
Methodology & Equations	**2.5**	Weak	■■■■■
Statistical Analysis	**1.5**	Very Weak	■■■■■
Results & Evidence	**2.5**	Weak	■■■■■
Discussion & Conclusions	**1.5**	Very Weak	■■■■■
Figures & Tables	**3.5**	Good	■■■■■
Scientific Writing	**3.0**	Acceptable	■■■■■
References & Citations	**2.5**	Weak	■■■■■
Ethics & Transparency	**1.5**	Very Weak	■■■■■

Detailed Agent Reviews

TitleAbstractKeywordsReviewer

Title, Abstract, Keywords

Score: **2.5 / 5.0** | Confidence: 90%

Weaknesses:

- [TITLE] The title is not specific and does not reflect the manuscript's contribution.
- [ABSTRACT] The abstract is missing and does not provide any structured content.
- [KEYWORDS] The keywords are insufficient and duplicate title words.

Major Comments:

- [TITLE] The title is not a valid manuscript title but appears to be a journal page header. It lacks specificity and informativeness.
- [ABSTRACT] The abstract is not detected, which is critical for standalone readability and structured reporting.
- [KEYWORDS] Only one keyword is provided, which is insufficient for discoverability and does not include domain-specific terms.

Minor Comments:

- [TITLE] The title should be revised to reflect the actual contribution of the manuscript.
- [KEYWORDS] Additional domain-specific keywords should be included to enhance discoverability.

Recommendations:

- [TITLE] Revise the title to be specific and informative, reflecting the actual contribution of the manuscript.
- [ABSTRACT] Provide a complete abstract with structured sections: Background, Objective, Methods, Results, and Conclusions.
- [KEYWORDS] Include 4-8 domain-specific keywords that are not duplicated from the title.

StructureReviewer

Manuscript structure and logical coherence

Score: **2.0 / 5.0** | Confidence: 80%

Strengths:

- Clear section titles and organization
- Presence of key sections like methodology, results, and conclusions

Weaknesses:

- Missing abstract and acknowledgments
- Poor logical flow between sections
- Lack of clear research problem and hypothesis
- Inconsistent section balance
- Missing transitions between sections

Major Comments:

- The manuscript lacks a clear research problem and hypothesis, which is essential for a structured scientific paper.
- The logical flow between sections is not coherent, and the structure does not follow the standard Introduction → Literature → Methods → Results → Discussion → Conclusions format.
- Missing abstract and acknowledgments are significant omissions for a Q1 journal.
- The section balance is uneven, with some sections being disproportionately long or absent.
- The conclusions do not clearly address the stated objectives, which undermines the coherence of the manuscript.

Minor Comments:

- Some sections contain incomplete or fragmented content, such as the introduction and results sections.

- The references section is missing or incomplete.
- The experiments and discussion sections are not well integrated with the rest of the manuscript.

Recommendations:

- Add an abstract and acknowledgments section to meet journal standards.
- Revise the structure to follow the standard scientific format: Introduction → Literature → Methods → Results → Discussion → Conclusions.
- Clarify the research problem and hypothesis in the introduction.
- Ensure that objectives are clearly stated and addressed in the conclusions.
- Improve the coherence and transitions between sections to enhance logical flow.

MethodologyReviewer

Methodology, experimental design, equations — holistic review

Score: **2.5 / 5.0** | Confidence: 55%

Strengths:

- The paper presents a computational model for time series forecasting.

Weaknesses:

- The methodology is not clearly described, making it difficult to understand the proposed model.
- The experimental setup and dataset are not described, making replication impossible.
- The results and discussion are not well-connected to the experiments, and the conclusion is vague.

Major Comments:

- The methodology is not clearly described, and the optimization model is not well-defined.
- The experimental setup, dataset, and evaluation metrics are not described, making replication impossible.
- The results and discussion are not well-connected to the experiments, and the conclusion is vague.

Minor Comments:

- Equations are not clearly defined or explained in the context of the optimization model.

Recommendations:

- Provide a clear problem statement and motivation for the proposed model.
- Describe the experimental setup, dataset, and evaluation metrics in detail.
- Clarify the equations and their role in the optimization model.
- Ensure that the results and discussion are well-connected to the experiments.

StatisticsReviewer

Statistical methods, tests, and reporting quality

Score: **1.5 / 5.0** | Confidence: 80%

Strengths:

- Use of Kruskal-Wallis and Wilcoxon/rank-sum post-hoc tests for non-parametric comparisons
- Reporting of RMSE, MSE, and coefficient of variation
- Use of 30 independent runs (if justified)

Weaknesses:

- Lack of effect sizes and confidence intervals
- No justification for sample size
- No multiple comparison corrections
- No verification of test assumptions
- No reproducibility details
- No distinction between statistical and practical significance

Major Comments:

- The manuscript fails to report effect sizes, confidence intervals, and reproducibility details, which are critical for assessing the validity and generalizability of results.
- The use of 30 independent runs is not justified, and the manuscript does not provide a rationale for this choice.
- The manuscript does not verify the assumptions of the statistical tests used, which is essential for the validity of non-parametric methods.

Minor Comments:

- The manuscript should clarify the interpretation of the coefficient of variation and RMSE/MSE metrics.
- The use of Wilcoxon/rank-sum post-hoc tests should be accompanied by proper multiple comparison corrections (e.g., Holm).

Recommendations:

- Report effect sizes (e.g., Cohen's d, eta-squared) and confidence intervals for all comparisons.
- Provide a justification for the sample size (30 runs) and include a power analysis if applicable.
- Apply multiple comparison corrections (e.g., Holm) for post-hoc tests.
- Verify the assumptions of the statistical tests (normality, independence, homoscedasticity) and report relevant diagnostic checks.
- Clarify the distinction between statistical and practical significance.
- Include reproducibility details such as random seeds, run configurations, and variance measures.

FiguresTablesEquationsReviewer

Figures, tables, equations — quality and coherence with text

Score: **3.5 / 5.0** | Confidence: 62%

Strengths:

- 7 figure references were detected in the manuscript.
- 19 sequentially numbered equations were detected, suggesting a substantial mathematical formulation.

Weaknesses:

- The model response could not be used reliably, so this fallback verifies presence and numbering but cannot judge visual quality or table body completeness.

Minor Comments:

- The model response could not be used reliably, so this fallback verifies presence and numbering but cannot judge visual quality or table body completeness.

Recommendations:

- Manually inspect captions, axes, units, table headers, and variable definitions in the final PDF.

ResultsReviewer

Results validity and empirical support

Score: **2.5 / 5.0** | Confidence: 80%

Strengths:

- MSE, RMSE, CV, and baseline comparisons are reported.

Weaknesses:

- Results do not clearly answer the stated research objectives.
- The results are not consistently presented across text, figures, and tables.
- The presentation of results is mixed with interpretation and unsupported claims.

Major Comments:

- The results section fails to clearly address the stated research objectives, making it difficult to assess the contribution of the study.

Minor Comments:

- The equation for MSE is incomplete and not properly formatted.
- The results are not consistently tied to figures or tables.

Recommendations:

- Clarify how the results answer the stated research objectives.
- Ensure that all quantitative results are fully detailed and properly formatted.
- Separate factual results from interpretations and unsupported claims.
- Ensure consistency in the presentation of results across text, figures, and tables.

DiscussionConclusionsReviewer

Discussion depth and conclusions validity

Score: **1.5 / 5.0** | Confidence: 80%

Strengths:

- Some quantitative results are presented in the conclusions section
- The manuscript includes references to prior work

Weaknesses:

- Discussion section lacks interpretation of results in the context of existing literature
- No clear limitations of the study are stated
- Conclusions are not directly derived from results
- No specific hypothesis is addressed in the conclusions
- Future work is not clearly specified or justified

Major Comments:

- The Discussion section is incomplete and lacks critical analysis of results in the context of existing literature.
- The Conclusions section does not clearly answer the original research question or hypothesis.
- The manuscript lacks a clear statement of the study's limitations.
- Future work is not well-defined or justified based on the results.

Minor Comments:

- The conclusions section contains some quantitative results but lacks clarity and structure.
- The references provided in the Discussion section are not properly integrated into the discussion of findings.

Recommendations:

- Revise the Discussion section to interpret results in the context of existing literature and address unexpected findings.
- Clarify the limitations of the study and discuss their implications.
- Ensure that the Conclusions section directly derives from the results and answers the original hypothesis.
- Specify future work directions with concrete and justified suggestions based on the findings.
- Structure the Conclusions section to avoid overgeneralization and ensure clarity.

WritingReviewer

Scientific writing quality, clarity, and style

Score: **3.0 / 5.0** | Confidence: 55%

Strengths:

- The manuscript contains enough extracted text for a representative writing review.
- Technical terminology and quantitative reporting signals are present in the extracted text.

Weaknesses:

- The model returned malformed JSON for writing review, so this fallback cannot provide sentence-level editing comments.

Minor Comments:

- The model returned malformed JSON for writing review, so this fallback cannot provide sentence-level editing comments.

Recommendations:

- Review representative paragraphs manually for transition quality, undefined acronyms, and overly long sentences.
- Ensure technical terms are defined on first use and used consistently across sections.

ReferencesReviewer

References quality, recency, and relevance

Score: **2.5 / 5.0** | Confidence: 80%

Strengths:

- Citations are consistent in style (IEEE format).
- Some recent references (2020-2023) are included.

Weaknesses:

- Lack of seminal foundational works in the field.
- Missing key areas such as time series analysis and neural network basics.
- Some references appear to be incorrectly cited or formatted (e.g., [63] has a DOI that does not match the reference text).

Major Comments:

- The references section lacks seminal foundational works, which is critical for establishing the context and novelty of the research.
- The absence of key areas such as time series analysis and neural network basics indicates a significant gap in the literature review.
- Some references appear to be incorrectly cited or formatted, which may affect the credibility of the manuscript.

Minor Comments:

- The citation style is consistent, which is a positive aspect.
- The inclusion of some recent references (2020-2023) is a strength, but more recent works (2023) are needed for fast-moving fields.

Recommendations:

- Include seminal foundational works in the field to provide context.
- Add references to key areas such as time series analysis and neural network basics.
- Verify the accuracy of all citations, especially those with DOI mismatches.

EthicsLimitationsReviewer

Research ethics, limitations, and transparency

Score: **1.5 / 5.0** | Confidence: 80%

Strengths:

- The manuscript presents a novel optimization model for time series forecasting, which is a significant contribution to the field.

Weaknesses:

- The manuscript lacks comprehensive reporting on ethical considerations, which is critical for ensuring the integrity and validity of the research.
- Limitations are not explicitly stated or discussed, which undermines the credibility of the conclusions.
- The reproducibility of the results is not addressed, which is essential for scientific validation.

Major Comments:

- The manuscript fails to report ethical approval for human or animal studies, which is a fundamental requirement for research involving such subjects.
- The absence of a data availability statement compromises the transparency and reproducibility of the research.
- The lack of conflict of interest declaration and funding disclosure raises concerns about the integrity and objectivity of the research.

Minor Comments:

- The manuscript does not state the author contributions, which is important for clarity in collaborative research.

Recommendations:

- The authors are strongly advised to include a statement on ethical approval for any human or animal studies conducted.
- A data availability statement should be included to ensure transparency and reproducibility.
- The authors should declare any potential conflicts of interest and disclose all funding sources.
- Author contributions should be clearly stated to enhance accountability and clarity.
- The manuscript should explicitly discuss the limitations of the study, including their impact on the conclusions and generalizability of the findings.

Consolidated Major Comments

These issues must be addressed before the manuscript can be accepted.

[1] [TITLE] The title is not a valid manuscript title but appears to be a journal page header. It lacks specificity and informativeness.

[2] [ABSTRACT] The abstract is not detected, which is critical for standalone readability and structured reporting.

[3] [KEYWORDS] Only one keyword is provided, which is insufficient for discoverability and does not include domain-specific terms.

[4] The manuscript lacks a clear research problem and hypothesis, which is essential for a structured scientific paper.

[5] The logical flow between sections is not coherent, and the structure does not follow the standard Introduction → Literature → Methods → Results → Discussion → Conclusions format.

[6] Missing abstract and acknowledgments are significant omissions for a Q1 journal.

[7] The section balance is uneven, with some sections being disproportionately long or absent.

[8] The conclusions do not clearly address the stated objectives, which undermines the coherence of the manuscript.

[9] The methodology is not clearly described, and the optimization model is not well-defined.

[10] The experimental setup, dataset, and evaluation metrics are not described, making replication impossible.

[11] The results and discussion are not well-connected to the experiments, and the conclusion is vague.

[12] The manuscript fails to report effect sizes, confidence intervals, and reproducibility details, which are critical for assessing the validity and generalizability of results.

Consolidated Minor Comments

These are suggestions that would improve the manuscript quality.

[1] [TITLE] The title should be revised to reflect the actual contribution of the manuscript.

[2] [KEYWORDS] Additional domain-specific keywords should be included to enhance discoverability.

[3] Some sections contain incomplete or fragmented content, such as the introduction and results sections.

[4] The references section is missing or incomplete.

[5] The experiments and discussion sections are not well integrated with the rest of the manuscript.

[6] Equations are not clearly defined or explained in the context of the optimization model.

[7] The manuscript should clarify the interpretation of the coefficient of variation and RMSE/MSE metrics.

[8] The use of Wilcoxon/rank-sum post-hoc tests should be accompanied by proper multiple comparison corrections (e.g., Holm).

[9] The model response could not be used reliably, so this fallback verifies presence and numbering but cannot judge visual quality or table body completeness.

[10] The equation for MSE is incomplete and not properly formatted.

[11] The results are not consistently tied to figures or tables.

[12] The conclusions section contains some quantitative results but lacks clarity and structure.

Specific Improvement Recommendations

1. [TITLE] Revise the title to be specific and informative, reflecting the actual contribution of the manuscript.
2. [ABSTRACT] Provide a complete abstract with structured sections: Background, Objective, Methods, Results, and Conclusions.
3. [KEYWORDS] Include 4-8 domain-specific keywords that are not duplicated from the title.
4. Add an abstract and acknowledgments section to meet journal standards.

5. Revise the structure to follow the standard scientific format: Introduction → Literature → Methods → Results → Discussion → Conclusions.
6. Clarify the research problem and hypothesis in the introduction.
7. Ensure that objectives are clearly stated and addressed in the conclusions.
8. Improve the coherence and transitions between sections to enhance logical flow.
9. Provide a clear problem statement and motivation for the proposed model.
10. Describe the experimental setup, dataset, and evaluation metrics in detail.
11. Clarify the equations and their role in the optimization model.
12. Ensure that the results and discussion are well-connected to the experiments.
